

# Supercharge Your Computer Vision Journey with the Intel® Geti™ Platform

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**Develop custom AI models at scale. Empower your team to streamline data labeling model, training, and optimization across the AI development process.**

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## The Business Imperative

In today's challenging business landscape, adopting computer vision technologies is essential for organizations to innovate faster, stay ahead of the competition, and enhance operations. This is the case across almost every market sector:

- Manufacturers want automated quality assurance solutions capable of detecting defects during the production process, rather than discovering them once the product is finished or shipped to customers.
- Healthcare professionals seek to diagnose patients more quickly and accurately while at the same time ensuring the security of confidential medical data.
- City managers are looking for better ways to monitor and control traffic flow—to improve quality of life for their citizens, reduce carbon emissions, and meet sustainability goals.
- Retailers are hungry for solutions that will help them manage a volatile supply chain, overcome labor shortages, track inventory more efficiently, and reduce financial loss due to shrinkage.

## Computer Vision Leads to New Opportunities

AI, computer vision, and deep learning offer the greatest opportunity for digital transformation and the most promise to revolutionize entire industries. By leveraging visual data, organizations can access an unprecedented source of rich, real-time information—a modern-day demonstration of the age-old adage that a picture is worth a thousand words.

The opportunities are enormous. Companies that delay in adopting these technologies may fall behind. But diving headlong into AI development can result in solutions that fail to meet expectations, leading to costly remediation, damage to a company's reputation, and loss of stakeholder confidence. In addition, creating scalable computer vision applications is a complicated and time-consuming task.

## The Challenges in Developing Computer Vision Solutions

Historically, developers have faced multiple barriers to building and implementing computer vision solutions. To begin with, in most cases an existing AI model pre-trained on a large data set to solve one problem likely won't be sufficient to solve a similar problem.

In part this is because pre-trained models are created using publicly available data or data from a simulated environment. To enable a specific use case, and ensure that the model performs its intended function when deployed in the user's environment, AI models may require custom training based on proprietary data.

Unfortunately, custom-training computer vision models is challenging. It's a time-consuming and resource-intensive process to train custom models that can be successfully deployed as scalable solutions. Typically, large data sets are required to train an AI model to an acceptable level of performance. In some cases, simply sourcing enough visual data can be difficult, as thousands or tens of thousands of images may be required.

In many computer vision tasks, the involvement of a domain expert is essential if the AI model is to succeed. For example, in medical research applications, a developer or data scientist will simply lack the background to distinguish a cancer cell from a healthy one. They will need an oncology specialist to help with annotation, training, and retraining to build an effective diagnostic model.

Domain experts are also needed to help with other factors like ethics and data privacy. Domain experts and compliance teams must be involved in the AI training process to eliminate bias from computer vision models and ensure that privacy regulations are being followed to the letter.

But nontechnical team members are busy with their day-to-day work, and consequently are unable to devote weeks or months to AI model training. And as a rule, they will be unfamiliar with the complex tools and platforms used by data scientists and AI specialists. As a result, teams are often siloed, making collaboration difficult.

In most cases the work of training AI models is not going to be a "one and done" effort, because models need to be updated post-deployment. AI-powered solutions are built to address real-world needs—but those needs may change from month to month, or even week to week.

New product types are brought into production, retail businesses expand into new regions with different merchandise and consumers, lighting conditions in factories change. And so suddenly, an AI model needs updating. If the cost and difficulty of training that model was high in the first place, retraining starts to look impracticable and unattractive.

Decision-makers may balk at going through the ordeal of model training all over again. An inflexible AI model can thus begin to constrain business and operational decisions—a far cry from the agility that the computer vision solution was supposed to provide.

Finally, there are system-wide architectural, infrastructure, and deployment challenges to consider.

For computer vision solutions to succeed at scale, their underlying elements—model development features, collaborative work platforms, testing and debugging suites, inferencing engines, and performant computing—must all work together. But there is a distinct lack of standardized interfaces and comprehensive platforms in the current AI ecosystem. This places a burden on technical teams and creates a drag on productivity. Time that could be used to bring new ideas to light is instead spent maintaining and integrating the current infrastructure. AI experts essentially become tech support for their own solutions, spending hours trying to make incompatible parts and pieces fit together.

Developers must account for the challenges they will face once their AI models are deployed as well. They may need to optimize models for diverse hardware and ensure efficient inferencing on edge devices with limited resources.

These considerations are crucial to building scalable solutions that perform in the real world, yet are often difficult to get right during the development phase—a major reason why so many AI projects never make it past proof-of-concept. Up to now, the AI development process has been full of such inefficiencies and frustrations, leading organizational leaders to wonder why there isn't a better way to build computer vision solutions.

Overcoming these challenges and creating new opportunities are the inspirations behind the Intel® Geti™ software platform.

## Build Models Rapidly and Fast-Track Innovation with Intel® Geti™ Platform

[The Intel® Geti™ platform](#) is a breakthrough in AI model training: a comprehensive development platform for building computer vision models—encompassing everything from data upload and data labeling, to training, optimization, and achieving a working model.

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According to a Gartner® CIO survey, "17% to 25% of organizations planned to deploy AI within the next 12 months every year from 2019 to 2024, while the annual growth of production deployments was only 2% to 5%." Gartner also noted, "On average, 54% of projects make it from pilot to production."<sup>1</sup>

Designed for collaboration and ease of use, the platform enables businesses to build real-world computer vision applications more quickly and efficiently.

The Intel Geti platform makes it possible for data scientists and AI developers to create sophisticated computer vision models for use cases as varied as detection, classification, segmentation, anomaly detection, and task chaining with less effort. Data scientists and AI developers benefit from powerful, fine-grained controls to help perform more advanced tasks as well. They can:

- Fine-tune learning parameters including rate, batch size, and number of epochs
- Define active learning parameters like the number of annotations required for initial training, to make predictions, and to retrain
- Adjust parameters for rotation, tiling, scaling, and other augmentations
- Define pre- and post-processing for image sizes, soft thresholds, blur strength

And the [Intel® Geti™ SDK](#) makes it even easier for AI developers to programmatically interact with the platform and perform these functions from their own development environment. At the same time, domain experts and business users benefit from a platform so intuitive and user-friendly that even non-developers can train an AI model independently.

The Intel Geti platform is a true collaboration solution that supports distributed work and makes it easier for domain experts in different geographies to contribute seamlessly.

The Intel Geti platform helps with big-picture concerns relevant to business leaders, security teams, and compliance groups as well. Organizations benefit from a platform designed to facilitate cross-functional collaboration and break down business siloes. To conform with stringent cybersecurity and data privacy requirements, it can be deployed at the edge, on-premises, or on a virtual machine in the AWS cloud.

The vision behind the Intel Geti platform also helps the wider AI ecosystem win. Systems integrators (SIs), original equipment manufacturers (OEMs), and independent software vendors (ISVs) can all leverage the Intel Geti platform to go to market faster with the innovative and highly customized computer vision solutions their customers want.

OEMs can more easily incorporate computer vision features into their product offerings with a platform that puts AI model development within reach—even if they have limited developer resources. ISVs building computer vision solutions can leverage the Intel Geti platform to rapidly iterate custom AI models based on evolving customer needs—or even explore new service offerings such as custom computer vision model as a service. And SIs wanting to offer end-to-end, market-ready solutions that can deliver greater model development agility to their customers and better integrate AI training and model development into their solution pipeline.

## Save Time in Model Development with Generative AI, Active Learning, and AI Assistance

The Intel Geti platform includes a rich feature for building models with less effort.

**Active learning** is the cornerstone of the Intel Geti approach to AI model training. It is a machine learning methodology in which an information source or user is queried directly by the learning algorithm and accelerates the work of accurately labeling training data. With active learning, the algorithm chooses data points for labeling, strategically selecting the samples from the data set that will allow it to learn in the most advantageous way possible.

It helps with the overall efficiency of model training and reduces the total data needed to reach desired model quality. Instead of having to train an AI model on thousands of images or terabytes of video data, only 20 to 30 images or a short video clip are needed to train a first-round model. This shortens the process of getting a model to the testing and refinement phase, and then iteratively improving the model quality to the desired level.

**AI-assisted workflows** make the work of model training easier. Laborious data labeling is simplified with the help of AI features. The custom model, as it is trained, facilitates labeling data as the domain expert provides feedback during the active learning training cycle.

**Generative AI-assisted annotation** allows users to mark up a few examples in an image and then let the algorithm fill in the rest. The AI won't get it right 100% of the time, but in general, it will be far easier for the model developer to simply correct a few mistakes than to label dozens of data points in each image individually.

The platform also leverages a lightweight version of the generative-AI Segment Anything Model (SAM) developed by [Meta AI](#) to enhance the labeling experience. Users can segment objects and create bounding boxes with a click. Annotation is as simple as hovering over an object in an image or frame to see an annotation preview, selecting the object to apply the mark, and fine-tuning the results by selecting areas to add or remove.

**Time-saving UI features** offer the same type of capabilities found in professional image-editing applications. The Intel Geti platform lets users knock out backgrounds and autodetect similar objects, and makes batch processing, image analysis, and labeling drag-and-drop simple.

It is designed to help domain and business users participate in the early stages of training without having to learn complex data science tools or drop into Python.



### Interactive Model Training

Get started annotating data with as little as 20-30 images; then let active learning help you teach the model as it learns.



### Multiple Computer Vision Tasks

Create models for AI tasks including classification, object detection, semantic segmentation, or anomaly detection.



### Task Chaining

Train your model into a multistep, smart application by chaining two or more tasks, without the need to write additional code.



### Smart Annotations

Expedite data annotation and easily segment images with professional drawing features like a pencil, polygon tool, and OpenCV GrabCut.



### Production-Ready Models

Output deep learning models in TensorFlow or PyTorch formats (where available) or as an optimized model for OpenVINO™ toolkit to run on Intel® architecture CPUs, GPUs, and VPUs.



### Hyperparameter Optimization

Refining hyperparameters is critical to the model's learning process. With built-in optimization, the Intel Geti platform makes a data scientist's job easier.



### Rotated Bounding Boxes

Support for rotated bounding boxes extends the training simplicity and accuracy to datasets with images that are not axis-aligned.



### Model Evaluation

Comprehensive statistics to assess your model's performance.

Intel® Geti™ platform core capabilities.

## Solve Architectural Challenges with the Intel® Automated Self-Checkout Reference Platform—from POC to Deployment at Scale

It's one thing to talk about the potential benefits of computer vision solutions. But it's quite another for retailers to achieve those benefits in practice and at scale. Consider automated self-checkout. The first step toward deployment would be to build a working proof-of-concept. But POCs built under curated conditions won't tell decision-makers much about how an AI model would fare in production—or what it will really cost to deploy at scale.

And it's difficult to determine in advance what hardware and what system architecture will be required to deploy a self-checkout solution. There are simply too many open questions. What video quality will be necessary to make the solution work? What is the minimum compute we'll need in-store? Do we deploy services on-device, or distributed across systems? How much latency is acceptable? How many video streams must be handled at a minimum throughput? And above all: What will the solution cost to deploy and maintain across the business, and is it worth that cost?

To help retailers develop AI solutions that scale, Intel developed the [Intel® Automated Self-Checkout Reference Platform](#), which offers practical support for solutions developers and software architects wanting to move from proof-of-concept to real-world deployment.

The reference platform offers benchmarking tools to help developers evaluate their computer vision workload on different hardware platforms. This lets them choose the right hardware to minimize cost per workload while still supporting their solution's intended use case. It also helps them make better choices about system and architectural elements.

It also helps business decision-makers understand the end-to-end deployment costs of a proposed solution. This lets them make informed decisions about whether the solution makes sense from an ROI standpoint at the beginning of the development process, rather than spending months on developing a POC only to discover that it won't be cost-effective to implement at scale.

The platform speeds time-to-market by leveraging well-known best practices, providing code blocks, offering helpful documentation in the form of videos and blog posts, and furnishing performance data to help developers bootstrap their product and project work.

Developers can easily integrate their own custom computer vision model into the reference system to prototype solutions that will work as intended in their specific use case. The Intel Automated Self-Checkout Reference Platform is a powerful tool for developers and businesses—and it's only the first of many reference platforms in Intel's open-source initiative for vision-enabled use cases with more to come ahead.



## Computer Vision Enables New Opportunities in Retail

Retail is a great example of the many ways computer vision supports a wide range of use cases. With a multitude of challenges—from staffing shortages and supply chain volatility to rising customer expectations and stiff economic headwinds—computer vision can be game-changing.

**Reduce shrinkage:** Self-checkout maybe the biggest operational headache for retailers. In 2022, retail shrink in the U.S. alone cost retailers an estimated \$112 billion.<sup>2</sup> With a problem of that magnitude, it's no wonder business leaders are eager for effective loss-prevention solutions.

Behavioral recognition algorithms can be put to use detecting SKU swaps by employees or shoppers, whether these are malicious or accidental. Self-checkout theft can be stopped before it happens by implementing barcodeless scanning to build a shrink-proof self-checkout kiosk that can't be tricked.

**Deliver frictionless customer experiences:** No one likes waiting in long lines. Automated self-checkout is the clear answer to speeding up the checkout process—but existing solutions have limitations. By leveraging computer vision, developers can create solutions that detect product SKUs with high accuracy, identify non-barcoded items quickly, and even recognize items placed in transparent bags.

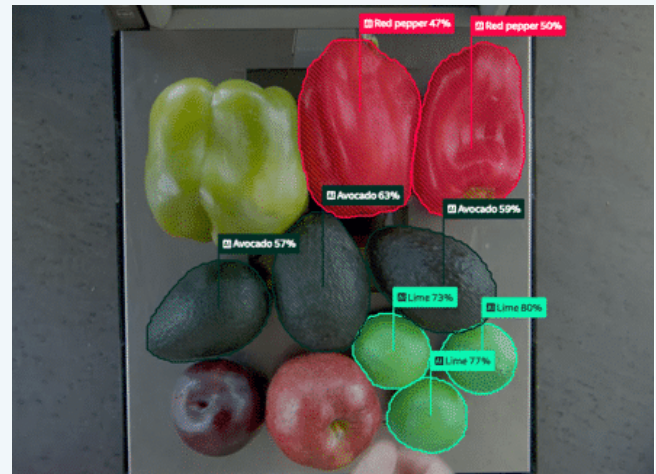
Vision-based solutions can alleviate customer bottlenecks and reduce wait times while sidestepping the frustrations of non-intuitive and temperamental self-checkout kiosks. By automating the checkout process, retailers also free up in-store personnel to pursue higher-value tasks or provide personalized service to customers who need it.

**Alleviate staffing shortages:** 7 in 10 surveyed retail executives called labor the number-one challenge.<sup>3</sup> Computer vision holds the potential to automate manual processes from inventory management and stock receiving to customer checkout and store monitoring.

The result improves the in-store experience for customers—and also for retail workers, who are often difficult for businesses to retain due to on-the-job frustrations.

**Improve inventory and supply chain management:** Unstable supply chains are a major issue for retailers, and staffing shortages have made it even harder to ensure on-shelf availability.

Computer vision product recognition can be used to automate inventory management and low-stock alerts, helping retailers keep track of what's on their shelves and what needs to be ordered immediately to avoid a stockout. In this way, retailers can prevent lost sales, improve the customer experience, and compete in an increasingly online world.



The Intel® Geti™ platform helps expedite data labeling and model building.

**Planogram optimization:** Inflation and economic uncertainty are widely expected to cause consumers to decrease spending. It's therefore more important than ever for retail businesses everywhere to improve profitability. Planogram optimization is a good way to make the most of every customer who enters a store, and computer vision can help in this effort by giving retailers detailed insights into product lifecycles and the in-store customer journey—even across thousands of stores and over widespread geographies.

The data acquisition and behavioral recognition capabilities of computer vision can help retailers gain insight into differences in customer profiles at various store locations, quantify ideal product placement, see which products are performing well, and understand shopping behavior to further optimize in-store layouts.

**Personalized digital signage:** Digital signage helps retailers increase sales and improve the customer experience. The behavioral detection and object recognition capabilities afforded by computer vision enable innovative digital signage solutions—personalized and optimized to boost sales and deliver engaging physical-digital experiences. These solutions can be used to show targeted messaging in real time. Messaging can also be localized across large chains via a centralized operations hub.

**Exceed customer expectations:** In general, computer vision can be used to make the brick-and-mortar experience more like online shopping: in other words, convenient, reliable, and efficient. Consumers don't necessarily avoid physical stores—in fact, many shoppers prefer them to online.<sup>4</sup> But people do dislike the inconveniences associated with in-person shopping. By using computer vision to deliver omnichannel experiences in-store, retailers can reimagine the retail experience and drive customer loyalty.

## Develop with the Intel® Geti™ Platform— Deploy with the Intel® Distribution of OpenVINO™ Toolkit—Integrate Workflows with the Intel® Geti™ SDK

The Intel Geti platform revolutionizes the first of two critical stages in the AI computer vision value chain: developing an AI model that solves a business problem.

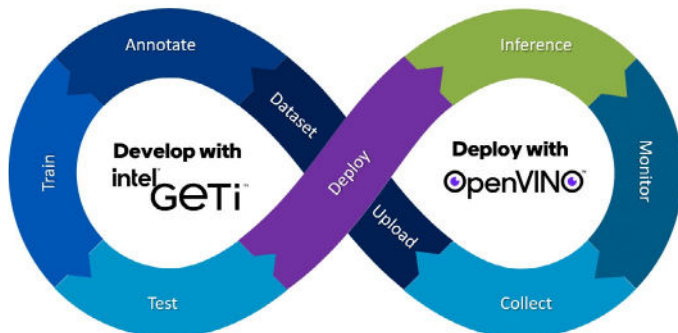
But once a custom AI model has been developed, it's time to consider the second stage: inferencing, in which the trained AI model is deployed in production to generate real-time analysis on new data.

The Intel Geti platform makes it possible for developers to prepare their models for deployment and inferencing in several ways. Built for flexibility, the platform can export production-ready deep learning models in several formats, including popular open-source frameworks such as [TensorFlow](#) (where available), [PyTorch](#), and [Intel® OpenVINO™](#) for Intel® CPUs, GPUs, and VPUs, and ONNX format.

Developers who choose to handle their inferencing workloads using OpenVINO can automatically optimize their models for deployment on virtually any Intel-based laptop, desktop, embedded PC, or server.

And with OpenVINO, developers gain the ability to test various architectural and compute considerations ahead of time, adding even more value for teams using resources like the Intel Automated Self-Checkout Reference Platform to maximize the possibility of their solution making it beyond proof-of-concept and into production.

Finally, the Intel Geti SDK helps teams bring everything together—enabling them to better integrate their OpenVINO and Intel Geti workflows, build deployment pipelines, and accelerate inferencing on various Intel hardware platforms. The one-two punch of the Intel Geti platform and Intel OpenVINO, seamlessly integrated by APIs using the versatile SDK, gives teams two major pieces of the puzzle required to create a powerful and efficient computer vision workflow.



Integrate workflows with the Intel® Geti™ SDK.

## Optimize and Deploy with Intel® OpenVINO™

Because AI inferencing is computationally demanding, a robust infrastructure capable of handling high throughputs is mandatory. And because many AI applications require inferencing to happen in real time and at the edge, there are

also demands for low latency and low power consumption. OpenVINO solves the problems involved in setting up inference pipelines in several ways:

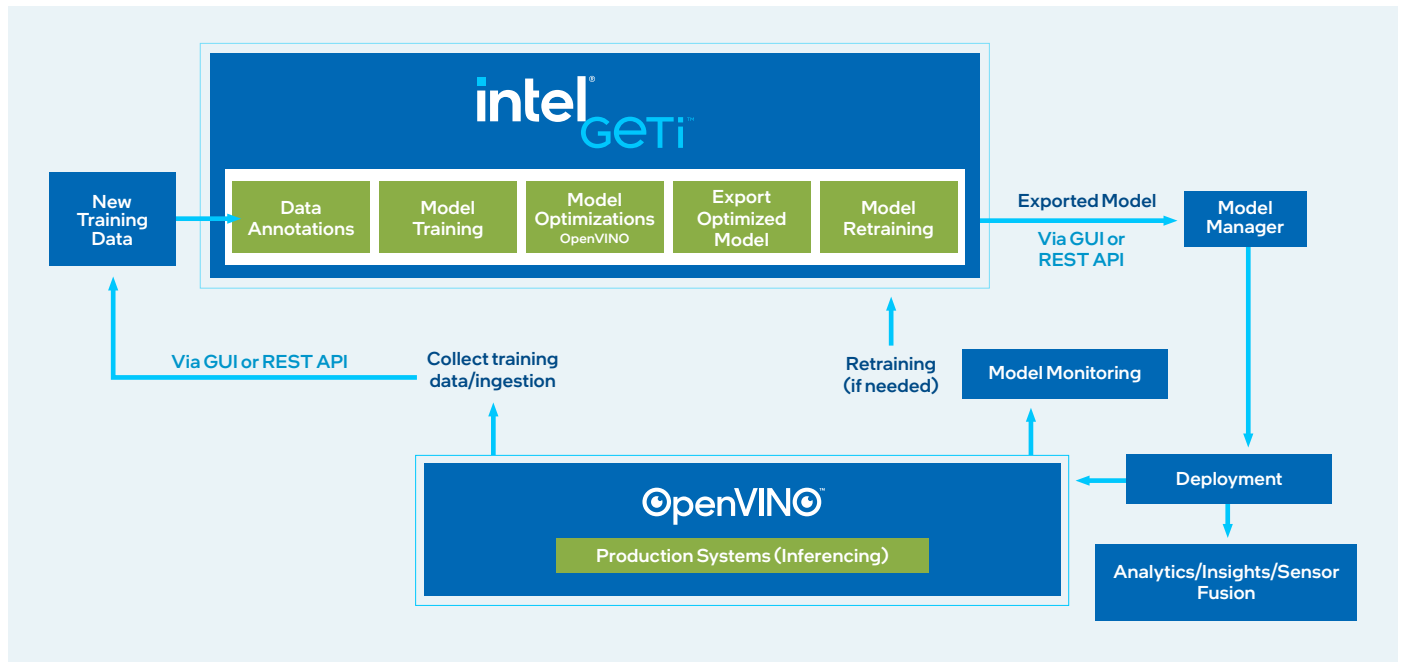
- Includes powerful optimization and quantization deployment tools. The OpenVINO Runtime API is full of toolkits to help developers, including an AI model optimizer and a neural network compression framework.
- Simplifies the integration of trained models into solutions that are deployed widely. AI inferencing workloads often need to run in different locations and on different device types.  
If developers had to build separate workloads for every different hardware type or operating system, it would be extremely cumbersome. Developers can build once and then use OpenVINO to create packages for whatever type of deployment is needed.
- Allows development teams to make architectural choices that work best for their use case—whether developers determine this through independent testing or by leveraging a reference platform. Inference packages can run as stand-alone applications or be deployed as containers using open-source or commercial platforms.
- Inferencing can run locally on devices, or as a microservice using [OpenVINO Model Server](#), an open-source system for serving machine learning models.
- Enables developers to optimize their models to run more effectively on the underlying hardware—and helps reduce the overall size of the model whether it is deployed locally or through a separate server. The real-world performance of AI models in production can also be monitored using OpenVINO post-deployment by setting up analytics pipelines.

OpenVINO allows developers to create real-time edge workflows, enabling key computer vision use cases at the edge and helping solutions architects reduce costly, large-scale data transfer by ensuring that data is consumed close to where it is produced.

## Streamline Your Workflow with the Intel® Geti™ SDK

The Intel Geti SDK is a Python package that lets developers interact with Intel Geti servers via REST APIs, making AI development workflows easier and more efficient. The SDK effectively helps teams mesh the capabilities of the Intel Geti platform and Intel OpenVINO—both during the AI model development stage and also post-deployment. It does this in several ways:

- Tools can be accessed within the Intel Geti platform to create projects from annotated data sets on disk, download whole projects and associated files, and create or upload projects from a previous download.
- Simplified project deployment for inferencing with OpenVINO. The SDK manages the work of pre- and post-processing in the background—agnostic to the computer vision processing task or architecture.



Build your end-to-end AI model training and deployment pipeline with the Intel® Geti™ platform and Intel® Distribution of OpenVINO™ toolkit.

- Enables teams to make smarter architectural choices for their AI solution. The SDK uses OpenVINO Runtime APIs in the background, wrapping AI models so inferencing can be done locally or as a microservice using remote servers (for example, using OpenVINO Model Server). Teams have the freedom and flexibility to deploy their AI model for inferencing in the most optimal way.
- Provides a CI/CD approach to AI solutions development, making it possible to move seamlessly between the model training and deployment stages. If there is a new variable in the deployment environment—be it a change in lighting conditions, a never-before-seen type of manufacturing defect, or a new product category—the SDK makes the work of AI model retraining and redeployment far easier.
- Enables developers to send new visual data to the Intel Geti platform for model retraining and then use the SDK to push the updated model back to OpenVINO for redeployment. Everything happens seamlessly in a single, integrated development and deployment pipeline.
- Includes a wealth of tutorial-style demos for developers, such as [Jupyter notebooks](#), so platform users can start taking advantage of its many capabilities right away.

## The Intel Ecosystem: Building for Tomorrow and Beyond

The Intel Geti platform, OpenVINO toolkit, and the Intel Geti SDK comprise a comprehensive computer vision model development and deployment system. In itself, that is enormously valuable to businesses. But these technologies

are also part of the wider ecosystem of Intel solutions and open-source projects, adding further value for businesses that leverage them to develop AI computer vision solutions.

The Intel hardware and software stack is ideal for computer vision tasks, especially at the edge. From high-performance CPUs and GPUs to edge server solutions and low-power processors for embedded applications, Intel offers a diverse range of options for computer vision processing. In addition, Intel® RealSense™ cameras provide a powerful suite of solutions for visual data acquisition.

Put another way, with Intel, developers can implement major portions of their AI computer vision solution—from model training, deployment, and software development to compute and sensors—and stay entirely within the same technology partner ecosystem. This means fewer integration challenges for development teams, and the operational advantage of working with core elements are already optimized for maximum efficiency.

Beyond the technology, Intel is committed to helping businesses, solutions providers, and system integrators capitalize on digital transformation opportunities more effectively, supporting this vision of digital transformation through its involvement with hundreds of open-source projects. Intel regularly collaborates with software engineers in the open-source community, provides code samples to developers, and supports and maintains a multitude of open-source initiatives.

By developing computer vision solutions within the Intel ecosystem, you can have greater confidence that your digital transformation journey is being built on a firm foundation—a family of integrated tools and technologies that will continue to be supported, refined, and improved in the years ahead.

## Transforming Industries—and the World—Through AI

Artificial intelligence has the potential to create positive change in nearly every sector and industry, from manufacturing, retail, and logistics to medicine, agriculture, and smart cities. Using AI, organizations have the opportunity to use resources more efficiently, optimize operations, gain insight into data as never before, and greatly simplify workflows and processes.

The Intel Geti platform empowers teams to realize this potential, allowing them to focus on deploying and consuming AI to solve real-world problems—instead of spending all their time and energy building it. With more time and fewer barriers, teams are free to create innovative, AI-powered solutions that deliver the benefits of digital transformation to their organizations and the wider world, helping build a more equitable, productive, and sustainable future for everyone.

### Intel® Geti™ Platform in Action

#### Showcase platform's potential

A representative example of the potential for rapid AI model development is with a proof of concept (POC) for an agricultural machinery manufacturer deploying computer vision for defect inspection. The manufacturer wanted to reduce human error in complex machinery assemblies by checking the quality of dozens of individual parts and verifying that the numerous steps in their production process were being executed in the correct sequence and in the right way.

The manufacturer had a quality assurance program in place, but it was entirely dependent on skilled engineers manually collecting and inspecting parts, making it extremely time-consuming. Although the manufacturer's technicians were well-trained, the assembly workflow was so complex they needed to complete error-free, multi-step processes several times per day, introducing the unavoidable possibility of human error. This production bottleneck stymied the company's need to scale operations without sacrificing quality or increasing headcount. What they really wanted

was a computer vision defect inspection solution—but were challenged by the difficulty and inefficiency of the AI training process.

Using the Intel Geti platform, the manufacturer's consultancy developed a computer vision-based defect inspection solution. Even though the solution required approximately 20 separate AI models, the developers were able to train the AI models using hundreds of images instead of thousands. This helped reduce training time, eliminate false positives, and minimize data imbalances. Furthermore, the platform was so user-friendly that the manufacturer's domain experts were able to take over part of the AI model training process, lending their expertise to make the models more accurate. The result was a complete AI-based defect detection system that accomplished the manufacturer's QA goals. The POC was such a success that the consulting firm envisions opportunities in other industries and use cases including predictive maintenance and worker safety.

## Take a deeper dive into the Intel platforms and tools that ramp up your computer vision development:

[Intel® Geti™ Platform](#) | [Intel® Geti™ SDK](#) | [Intel® Distribution of OpenVINO™ Toolkit](#)

[Intel® Automated Self-Checkout Retail Reference Implementation](#)



<sup>1</sup> Gartner®, Become an AI First Organization: 5 Critical AI Adoption Phases, By Svetlana Sicilar, et. al, October 13, 2023. GARTNER is a registered trademark and service mark of Gartner, Inc. and/or its affiliates in the U.S. and internationally and is used herein with permission. All right reserved.

Intel does not control or audit third-party data. You should consult other sources to evaluate accuracy.

<sup>2</sup> <https://www.businessinsider.com/retailers-lost-112-billion-to-inventory-shrink-in-2022>

<sup>3</sup> <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/consumer-business/us-consumer-business-retail-outlook.pdf>

<sup>4</sup> <https://chainstoreage.com/heres-what-customers-dont-about-stores>

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